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Overview of Traffic Flow Forecasting Techniques

Annarita Carianni · A. Gemma · Computer Science, Engineering ·

IEEE Open Journal of Intelligent Transportation… · 2025

TLDR A comprehensive **review** of traffic forecasting methods, bridging traditional **techniques** and innovative approaches driven by computational **intelligence** and abundant data, with special emphasis on **artificial intelligence**’s transformative role in enabling dynamic and accurate predictions. Expand

7 · IEEE · Save · Cite

AI-DRIVEN MIS APPLICATIONS IN ENVIRONMENTAL RISK MONITORING: A SYSTEMATIC REVIEW OF PREDICTIVE GEOGRAPHIC INFORMATION SYSTEMS

Subrato Sarker · Faria Jahan · Environmental Science, Computer Science ·

ASRC Procedia: Global Perspectives in Science and… · 1 January 2025

TLDR Findings reveal that AI **techniques**—particularly **machine learning** and **deep learning** models—significantly improve the predictive power of GIS platforms, with over 60% of the **review**ed studies reporting model accuracy above 85%. Expand

2 · Publisher · Save · Cite

Machine Learning Principles for Radiology Investigators.

S. Borstelmann · Medicine, Computer Science · Academic Radiology · 1 January 2020

TLDR Advanced statistical **machine learning techniques** including bootstrapping, bagging, boosting, decision trees, random **forest**, XGboost, and support vector **machine**s are **review**ed along with relevant examples from the radiology literature. Expand

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AI-Driven Handover Management and Load Balancing Optimization in Ultra-Dense 5G/6G Cellular Networks

Chaima Chabira · Ibraheem Shayea · Gulsaya Nurzhaubayeva · L. Aldasheva · D. Yedilkhan · Saule Amanzholova

Computer Science, Engineering · Technologies · 1 July 2025

TLDR A comprehensive **review** of handover **management** and load balancing optimization (LBO) in ultra-dense 5G and emerging 6G cellular networks and how recent advances in **artificial intelligence** (AI) are being applied to improve predictive handover decisions and enable real-time, adaptive load distribution is presented. Expand

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MACHINE LEARNING FOR BIOMARKER DISCOVERY IN METABOLOMICS: TECHNIQUES, APPLICATIONS, AND FUTURE DIRECTIONS

S. Yadav · Abhishek Kumar Gautam · Medicine, Computer Science

TLDR The main developments in AI-driven metabolomics are highlighted in this **review**, with particular emphasis on the use of ML models such as random **forest**s, support vector **machine**s, and **artificial** neural networks for disease diagnosis and biomarker discovery, as well as **deep learning** approaches for metabolite identification and multi-omics data integration. Expand

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A Survey on Enrolment Analysis via AI-Based Forecasting Models in Academic Institutions

Leow Xien Jing · Muhammed Basheer Jasser · S. Ajibade · Hui Na Chua · Richard T. K. Wong

Computer Science, Education · International Conference on Electrical, Control… · 27 August 2025

TLDR The findings show that neural networks, particularly Long Short-Term Memory (LSTM) models, outperform traditional statistical methods in capturing non-linear patterns and temporal dynamics. Expand

1 IEEE Save Cite

Artificial Intelligence-Based Models for Predicting Cardiovascular Events: A Review of Current Trends and Future Directions

A. Singh · Ms. Namita Srivastava · Medicine, Computer Science

TLDR A comprehensive analysis of current trends and future directions in AI-based models for predicting cardiovascular events, including **machine learning** algorithms, **deep learning** models, and ensemble methods, employed for cardiovascular risk prediction is provided. Expand

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Giving Cows a Digital Voice – AI-Enabled Bioacoustics and Smart Sensing in Precision Livestock Management

Mayuri Kate · Suresh Neethirajan · Computer Science, Agricultural and Food Sciences · Annals of Animal Science · 26 August 2025

TLDR A Hybrid Explainable Acoustic Multimodal (HEAM) model is proposed, which fuses spectrogram-based convolutional neural networks, interpretable decision trees, and natural language reasoning modules to generate transparent and actionable alerts for farmers. Expand

3 Publisher Save Cite

Intelligent flood forecasting and warning: a survey

Yue Zhang · Daiwei Pan · J. Van Griensven · Simon X. Yang · Bahram Gharabaghi

Environmental Science, Computer Science · **Intelligence** & Robotics · 28 June 2023

TLDR Three typical **machine learning** algorithms for flood forecasting are **reviewed**: supervised **learning**, unsupervised **learning**, and semi-supervised**Learning**; special attention is given to **deep learning** approaches due to their better performance in various prediction tasks. Expand

12 Publisher Save Cite

A Systematic Review of Intelligent Stress Detection in Higher Education: Technologies, Challenges and Future Directions

Manjit Kaur · Devender Kumar · Education, Psychology · 2025 Second International Conference on… · 4 December 2025

TLDR A systematic **review** of intelligent stress measurement methods based on data sources, which had 35 studies in which **Machine Learning** models, such as Random **Forest**, were reported to have as high as 95% accuracy on large datasets, while **deep learning** algorithms were able to beat traditional methods when using multimodal data. Expand

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